

Experiment 2: Implementation of Basic gates using universal gates.

Theory: A universal gate is a logic gate which can implement any Boolean function without the need to use any other type of logic gate. The NOR gate and NAND gate are universal gates. This means that you can create any logical Boolean expression using only NOR gates or only NAND gates.

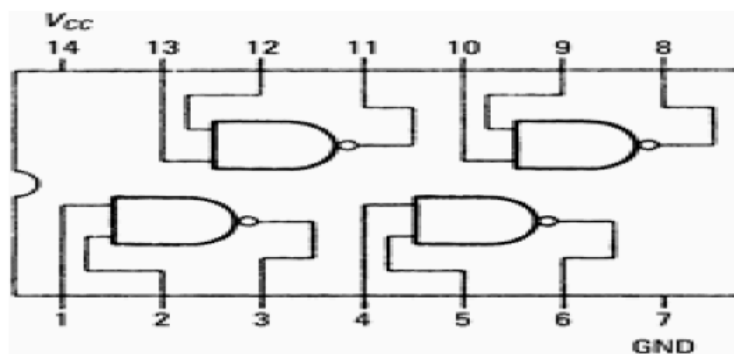
Apparatus Required:

1. Breadboard
2. Wire
3. Gates NAND (7400), NOR (7402) IC

Universal Gates

1. NAND gate: - Function of NAND gate is to give true output when one of the two provided input are false. In the remaining output is true case. Following table justifies the statement :

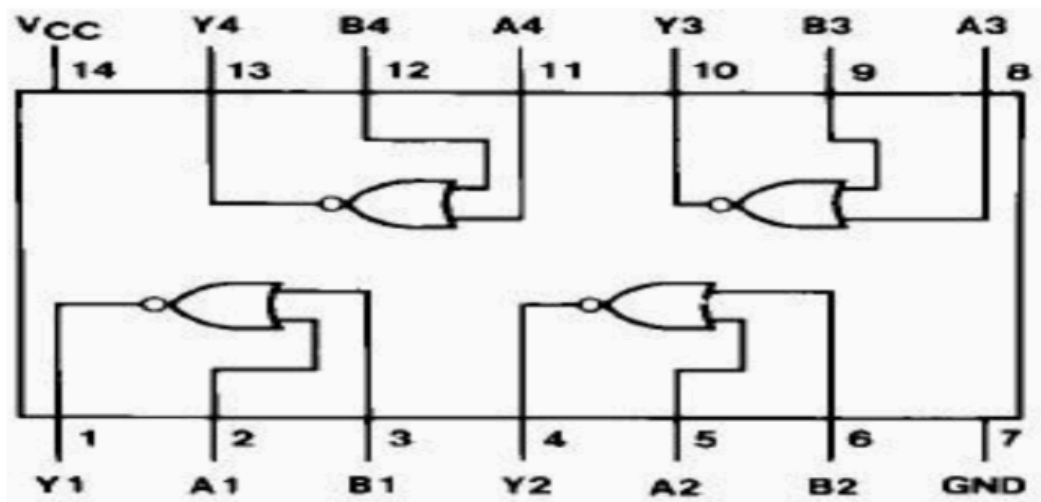
Input A	Input B	Output
1	1	0
1	0	1
0	1	1
0	0	1



IC 7400

2. NOR gate: - NOR gate gives the output true when both the two provided input are false. In all the other cases output remains false. Following table justifies the statement:

Input A	Input B	Output
1	1	0
1	0	0
0	1	0
0	0	1



IC 7402

We have seen the IC configuration of the NAND and NOR gate. Now we will be seeing the implementation of basic gates using the universal gates.

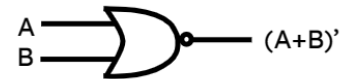
Basic gates using NAND gate



NOT	$A \rightarrow A'$	$(AA)' = A'$
AND	$A, B \rightarrow AB$	$((AB)')' = AB$
OR	$A, B \rightarrow A+B$	$(A'B')' = (A')' + (B')' = A+B$

We have seen how the NAND gate can be used to make all the three basic gates by using that alone. Now we will discuss the same in case of NOR gate

Basic gates using NOR gate



NOT	A NOR gate with input A and output A'.	A NOR gate with input A and output $(A+A)' = A'$.
OR	A NOR gate with inputs A and B and output $A+B$.	Two NOR gates. The first NOR gate has inputs A and B, output $(A+B)'$. The second NOR gate has inputs $(A+B)'$ and $(A+B)'$, output $((A+B)')' = A+B$.
AND	A NOR gate with inputs A and B and output AB.	Three NOR gates. The first NOR gate has input A and output A'. The second NOR gate has input B and output B'. The third NOR gate has inputs A' and B', output $(A'+B')' = (A')'.(B')' = AB$.

In practice, this is advantageous since NOR and NAND gates are economical and easier to fabricate than other logic gates. So much so that an AND gate is typically implemented as a NAND gate followed by an inverter.